Magnetic Circuits

Eddy Current Loss

(a) Eddy Current Loss

When a magnetic material (like the core of a transformer or motor) is subjected to a **time-varying magnetic flux**, circulating currents are induced within the material itself. These are called **eddy currents**.

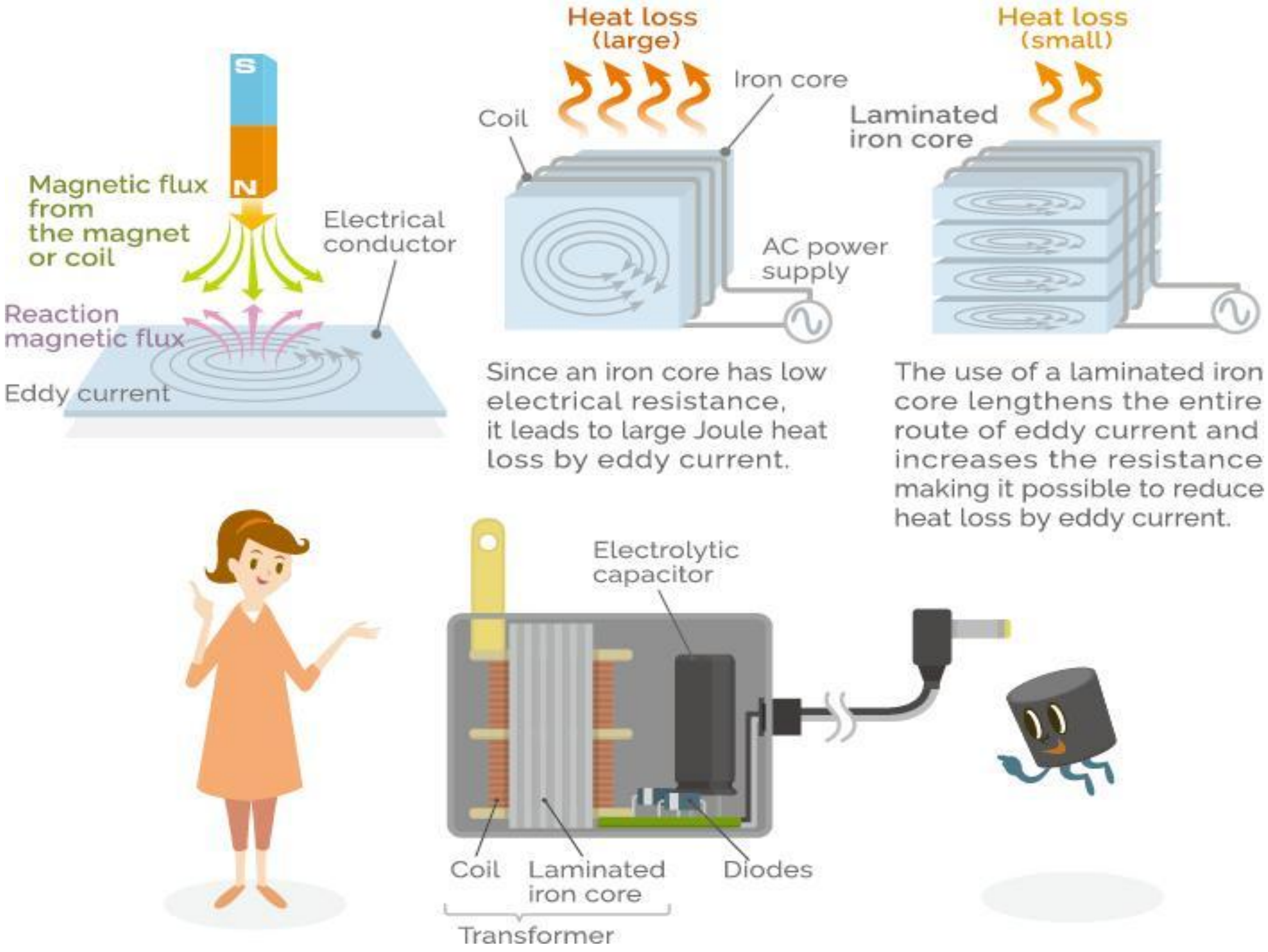
Why they occur

According to **Faraday's Law of Electromagnetic Induction**, a changing magnetic flux induces an EMF. In a bulk conductor, this EMF drives closed-loop currents inside the material.

Effects

- These circulating currents flow within the core material.
- Since the material has electrical resistance, they produce **I^2R losses**.
- Energy is dissipated as **heat**, reducing efficiency.

< Heat loss due to the iron core of a transformer and eddy current >



Mathematical Expression

Eddy current loss is approximately given by:

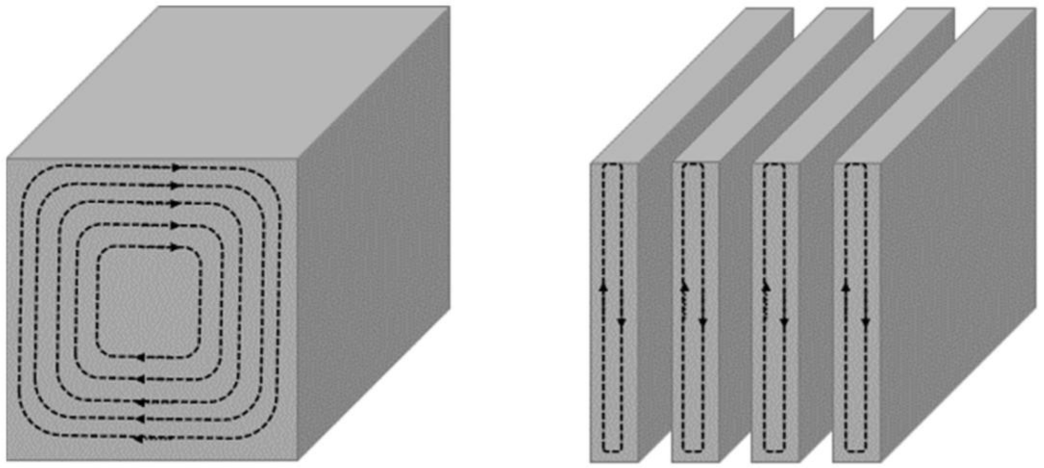
$$P_e \propto B_{max}^2 f^2 t^2$$

Where:

- B_{max} =maximum flux density
- f= frequency
- t= thickness of the core

Methods to Reduce Eddy Current Loss

- Use **laminated cores** (thin insulated sheets)
- Use materials with **high resistivity** (e.g., silicon steel)
- Reduce thickness of laminations



(a) Eddy current loop (b) Eddy current loop through lamination

Hysteresis Loss

Hysteresis Loss

Hysteresis loss occurs due to the repeated **magnetization and demagnetization** of a magnetic material when subjected to an alternating magnetic field.

Physical Explanation

Magnetic materials are made of **domains**. When an external magnetic field is applied:

- Domains align with the field.
- When the field reverses, domains must realign again.

This realignment is not perfectly reversible, leading to energy loss.

Hysteresis Loop

The relationship between magnetic field intensity (H) and flux density (B) forms a loop called the **Hysteresis Loop**.

•The **area of the loop** represents energy loss per cycle.

Mathematical Expression

$$P_h = \eta B_{max}^n f$$

Where:

•η= Steinmetz coefficient

•B_{max} =maximum flux density

•f= frequency

•$n \approx 1.6$(for most materials)

Characteristics

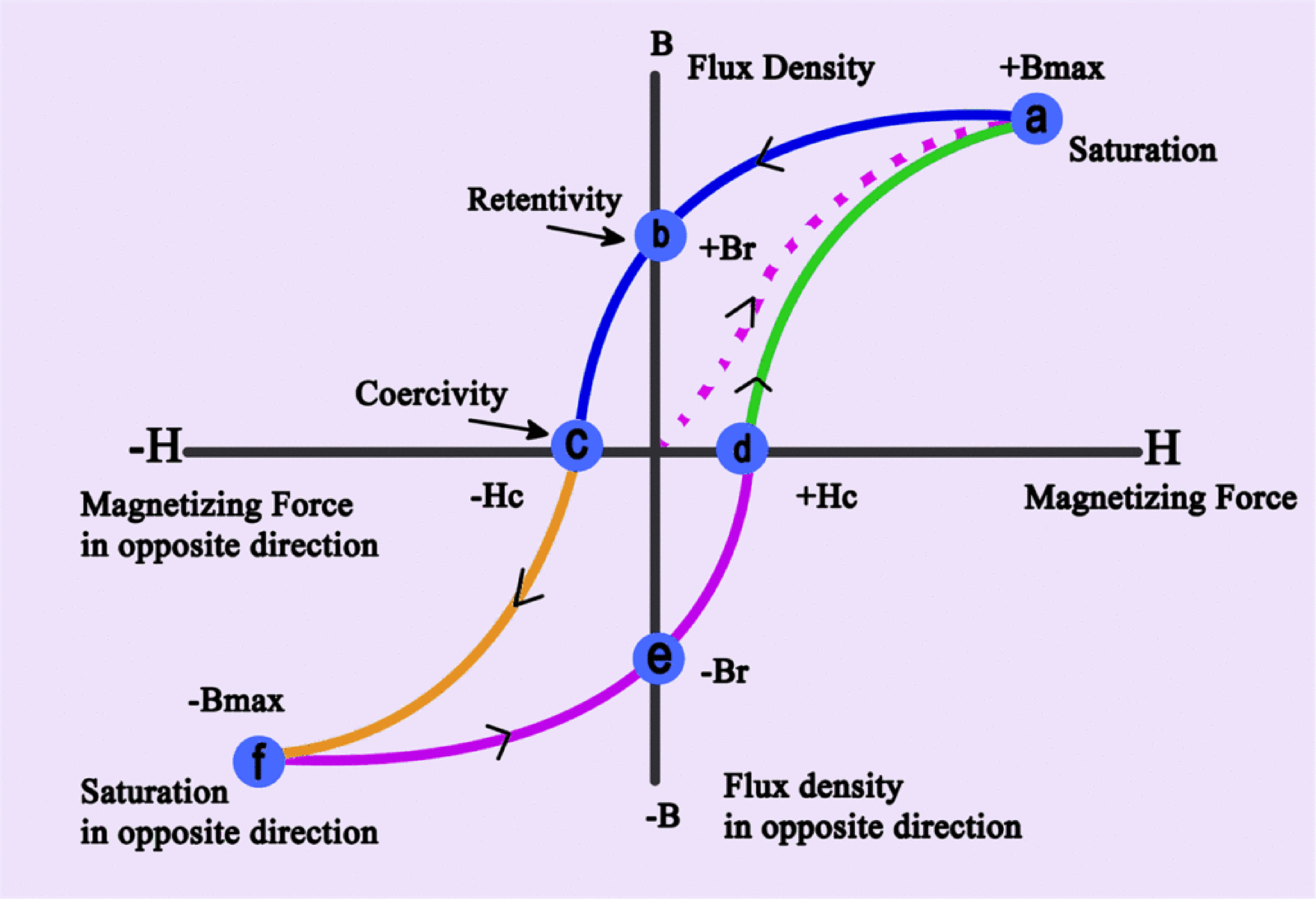
•Depends on material properties

•Increases with frequency

•Causes heating of the core

Reduction Methods

•Use materials with **narrow hysteresis loops** (e.g., silicon steel, ferrites)


B
Flux Density
+Bmax
a
Saturation
Retentivity
b
+Br
Coercivity
c
-H
d
H
-Hc
+Hc
Magnetizing Force
Magnetizing Force
in opposite direction
e
-Br
-Bmax
f
Saturation
in opposite direction
-B
Flux density
in opposite direction

The figure shown is the **B–H hysteresis loop** of a ferromagnetic material, which represents the relationship between the **magnetizing force** H applied to a magnetic material and the resulting **magnetic flux density** B produced in it during a complete cycle of magnetization. This loop is formed because the magnetic flux density does not follow the magnetizing force instantaneously; instead, it **lags behind** due to the internal molecular friction and domain realignment effects inside the magnetic material. This lagging phenomenon is known as **magnetic hysteresis**, and the closed curve obtained is called the hysteresis loop.

The horizontal axis of the graph represents the **magnetizing force** H, which is the magnetic field intensity applied to the material and is produced by current flowing through the exciting coil wound on the magnetic core. It is measured in ampere-turns per meter. The vertical axis represents the **magnetic flux density** B, which is the amount of magnetic flux developed per unit cross-sectional area of the material, measured in tesla. As the magnetizing current in the coil is varied from positive to negative and back again, the value of H changes, and the corresponding value of B traces the hysteresis loop.

To understand how the loop is formed, begin from the origin when the material is initially unmagnetized. As the magnetizing force H is increased in the positive direction, the flux density B rises along the **dotted initial magnetization curve** until it reaches point **a**, where the material attains **positive saturation**. At this point, nearly all magnetic domains inside the material are aligned in the direction of the applied field, and further increase in H produces negligible increase in B. The flux density here is the maximum positive flux density, denoted as $+B_{max}$.

When the magnetizing force is now reduced back to zero, the flux density does not return to zero. Instead, it follows the blue curve and reaches point **b**, where some magnetism remains in the material even though $H = 0$. This residual flux density is called the **retentivity** or **remanence**, denoted by $+B_r$. It represents the ability of the material to retain magnetization after the external magnetizing force is removed.

To reduce this residual magnetism to zero, a magnetizing force must be applied in the opposite direction. As negative H is applied, the flux density decreases and becomes zero at point **c**. The magnitude of reverse magnetizing force required to bring the residual flux density to zero is called the **coercive force** or **coercivity**, denoted by $-H_c$ .This indicates the resistance of the material to demagnetization.

If the negative magnetizing force is increased further, the material reaches point **f**, corresponding to **negative saturation**, where the flux density becomes $-B_{max}$ .At this stage, all domains are aligned in the opposite direction.

Now when the magnetizing force is reduced from negative saturation back toward zero, the flux density again lags and does not return along the original path. Instead, it follows the purple curve and reaches point **e**, where a residual negative flux density remains. This is the negative remanent flux density $-B_r$.
To bring this negative residual flux density back to zero, a positive magnetizing force must be applied. As H increases positively, the flux density becomes zero at point **d**, where the required positive coercive force is $+H_c$ .Further increase in positive magnetizing force returns the material to point **a**, completing the loop.
Thus, the closed path **a → b → c → f → e → d → a** forms the hysteresis loop. The loop exists because the magnetic domains in the material require energy to rotate and realign whenever the magnetic field changes direction. The **area enclosed by the hysteresis loop represents the hysteresis energy loss per cycle per unit volume of the magnetic material**. A larger loop area indicates greater hysteresis loss, while a narrower loop indicates lower hysteresis loss and better magnetic efficiency.
Physically, the "forces" involved in the graph are not mechanical forces but magnetic quantities. The **magnetizing force** H is the externally applied magnetic field generated by electric current through a coil, while the **flux density** B is the magnetic response of the material due to domain alignment. The loop is experimentally obtained by repeatedly magnetizing and demagnetizing a ferromagnetic specimen using alternating current and measuring the corresponding values of H and B.

Non-linear Magnetic Circuits

A magnetic circuit is said to be **non-linear** when the relationship between magnetic flux density (B) and magnetic field intensity (H) is not linear.

Cause of Non-linearity

This occurs due to **magnetic saturation** of the material.

Explanation

•Initially, $B \propto H$(linear region)

•As Hincreases, the material approaches saturation

•Beyond a certain point, large increases in Hproduce very small increases in B

This behavior is represented by the **B-H Curve**.

Implications

- Permeability (μ) is not constant
- Reluctance varies with flux
- Superposition does not apply
- Analysis becomes complex

Practical Consequences

- Magnetic devices (transformers, motors) must be designed to avoid deep saturation
- Non-linear analysis methods (graphical or numerical) are required

Eddy Current Loss

In a magnetic circuit, **eddy current loss** refers to the power dissipated in the magnetic core due to circulating currents induced within the core material itself when it is subjected to a time-varying magnetic flux. According to Faraday's law of electromagnetic induction, whenever the magnetic flux linking a conducting material changes with time, an electromotive force is induced in that material. Since magnetic cores are generally made of conducting ferromagnetic materials such as iron or steel, this induced emf drives circulating currents within the body of the core. These circulating currents are called eddy currents because they flow in closed loops resembling swirling eddies in water. As these currents pass through the electrical resistance of the core material, energy is converted into heat, leading to undesirable power loss and temperature rise in the magnetic circuit. Eddy current loss increases with the square of both the flux density and the supply frequency, and it also depends on the thickness of the core material. To minimize this loss, magnetic cores used in transformers, motors, and other AC magnetic devices are laminated into thin insulated sheets rather than made as a solid block. The lamination increases the resistance to the flow of eddy currents and significantly reduces their magnitude, thereby reducing the associated heating and energy loss.

Hysterisis Loss

Hysteresis loss is the energy dissipated in the magnetic material during each cycle of magnetization when the magnetic flux in the core repeatedly changes direction. Ferromagnetic materials are composed of numerous microscopic magnetic domains, each acting like a tiny magnet. When an external magnetizing force is applied, these domains align themselves in the direction of the magnetic field. However, due to internal molecular friction and domain resistance, the domains do not realign instantly or completely without expending energy. During every cycle of alternating magnetization, some energy is required to continuously reverse the orientation of these domains, and this energy is lost in the form of heat. The phenomenon responsible for this loss is known as magnetic hysteresis, which is represented graphically by the hysteresis loop of the material. The area enclosed by the hysteresis loop indicates the amount of energy lost per cycle per unit volume of the material. Hysteresis loss depends on the frequency of magnetization, the volume of the core, the maximum flux density, and the magnetic properties of the core material. Materials with narrow hysteresis loops, such as silicon steel, are preferred in magnetic circuits because they require less energy for repeated magnetization reversal and therefore exhibit lower hysteresis loss.